CLI-First: What an Agent Knows About an Interface Matters More Than Its Protocol

Chandra Irugalbandara
Leaf Monkey Labs
Colombo, Sri Lanka
chandra@leafmonkey.org

MCP, curated server (MCP-24)

```
get_trial_balance()
get_accounts()
create_reminder("Review spending: General Expenses", "2026-04-30")
create_reminder("Review spending: Fuel", "2026-04-30")
create_reminder("Review spending: Pharmacy", "2026-04-30")
… 21 more create_reminder calls
```

3 passes 26 tool calls 4.1k output tokens **11.7¢** 25 s

CLI with the command sheet, composed (CLI sheet/comp)

```
# call 1
salli accounts list --json > $TMPDIR/a.json 2>/dev/null
salli ledger trial-balance --json > $TMPDIR/tb.json
jq -r --slurpfile tb $TMPDIR/tb.json '.[]
  | select(.type=="expense") | [.id,.name,.currency,
    ($tb[0][.id]//"none")] | @tsv' $TMPDIR/a.json
# call 2
jq -r --slurpfile tb $TMPDIR/tb.json '.[]
  | select(.type=="expense" and .currency=="LKR")
  | select((($tb[0][.id]//"0")|tonumber)>20000)
  | .name' $TMPDIR/a.json |
while IFS= read -r n; do salli reminders add \
  "Review spending: $n" --due-date 2026-04-30 --json; done
```

3 passes 2 tool calls 0.9k output tokens **4.4¢** 48 s

Figure 1: The same task through two interfaces to one product. Asked to set a review reminder for every expense account over LKR 20,000, an agent with Salli's MCP server called a tool 26 times; an agent given the CLI's command sheet wrote a two-call script ending in one loop, produced a fifth of the output tokens and cost 62% less, but took twice as long and put 24 writes behind one tool call. This is the cell where composition saved most; over both repetition tasks the saving was 36–39%, and on chains CLI agents without the command sheet cost more than MCP (Section 6.5). Medians of three Sonnet runs on the large ledger. The left panel abridges an actual trace; the right reproduces one run's two commands, with line breaks added.

ABSTRACT

Coding agents now build and operate much of the software they touch, mostly from a terminal. We propose CLI-first development: a capability is done only when it works from the product's command-line interface, and that CLI is the reference client against which the other clients are checked. In a personal-finance product that adopted the rule, coverage fell behind until a dedicated phase restored it, and an agent's pass through the CLI found a safety control that held only on the web path. We then test a popular claim, that CLIs cost agents less than Model Context Protocol (MCP) servers, in 704 runs that hold the product fixed and vary the protocol, the agent's knowledge of the interface, and whether it may compose operations. What the agent knew mattered more than the protocol: agents that learned the CLI from its help cost 1.6 to 2 times as much as agents given the curated MCP server. With a compact command sheet, the CLI matched the server on chains of dependent steps for the stronger model and, composed, cost 36 to 39% less on two tasks that repeat an operation, at the price of coarser oversight. Follow-ups (1,440 runs) found that which parts of the sheet mattered depended on the model, and that on the product's rewritten version the composed CLI cost less than its MCP server, partly because that server had grown.

CCS CONCEPTS

• **Human-centered computing** → **Interaction paradigms**; • **Software and its engineering** → *Software verification and validation*; • **Computing methodologies** → *Intelligent agents.*

KEYWORDS

AI agents, command-line interfaces, Model Context Protocol, tool use, agent-computer interfaces, cost, oversight

1 INTRODUCTION

Agents are becoming a large share of the users of developer tools. Cloudflare reports that agents accounted for a quarter of the use of its command-line tool in March 2026, up from single digits a year earlier, and for 48% by late September [12]. Inside product teams the shift shows in version control: in Salli, the personal-finance product we study, 75% of commits carry a coding agent's co-authorship trailer. Those agents need to run the software they change, not just compile it.

We propose *CLI-first development*, a process with two rules: a capability is done only when it works from the product's command-line interface (CLI), and the CLI is the *reference client* that the product's other clients, including the web application and any Model Context Protocol (MCP) server, are checked against. The proposal makes two claims. One concerns process: a CLI held to these rules lets the agents that build a product exercise and verify all of it. We examine it in Salli's version history. The other concerns cost and is widely argued, while the few measurements disagree: that a CLI is cheaper for agents than an MCP server because MCP loads every tool definition into the model's context [51, 54], or more expensive because every call starts a process [36].

To test the cost claim we held one product fixed and varied only the interface its agents used (Figure 1), crossing three factors: the *protocol* (MCP tool calls or shell commands), what the agent *knows*

about the interface before it starts, and whether it may *compose* operations into one program. If the interface did not matter, every configuration would cost what Salli's curated MCP server costs, and every ratio we report would be 1. What the agent knew moved cost more than the protocol did. On chains, giving a CLI agent a command sheet instead of help lowered its cost relative to the MCP server from 1.6–2.0 to 0.9–1.2, a wider swing than the gap between the best CLI configuration and the server. Our thesis follows: the case for CLI-first rests on the agents that build and check a product, and for agents in general, what they are told about an interface matters more than which protocol carries it. We report five findings from the main experiment and two from follow-ups:

- **F1.** An agent that learned the CLI from its help cost 1.6 to 2 times as much as an agent given the curated MCP server's schemas (Section 6.5).
- **F2.** For the same operations, an MCP server generated from the CLI cost 1.3 to 1.8 times as much as the curated one; it also returned larger results, so the effect of catalogue size alone is not isolated (Section 6.6).
- **F3.** On chains of dependent steps, the CLI with a 1.1k-token command sheet and composition could not be told apart from the curated server for the stronger model, Sonnet, and cost 1.2 times as much for the weaker, Haiku (Section 6.7).
- **F4.** On two tasks that repeat an operation, the same configuration cost 36–39% less than the curated server for both models; Haiku saved only on bulk entry, and where one write was needed there was no consistent advantage (Section 6.8).
- **F5.** Composition moved the unit a person could review from single actions to whole scripts, and agents on both protocols, in four of eight configurations, reported success they had not achieved (Section 6.9).
- **F6.** Which parts of the command sheet mattered depended on the model: without its options the weaker model cost 1.24 times as much on chains and without its output shapes 1.10 times; the stronger model needed only the output shapes (1.19). A sheet four times larger cost no more under a warm shared cache but 1.27 to 1.59 times as much with a cold one (Section 7.1).
- **F7.** On the product as rewritten since, the composed CLI with a sheet cost 0.60 to 0.90 times as much as the product's MCP server on chains and 0.32 to 0.48 times on repetition tasks. Part of that came from the server doubling its catalogue: against its original 24 tools, the composed CLI was level for the weaker model on chains and cheaper elsewhere. Addressing accounts by code instead of id made no detectable difference (Sections 7.2 and 7.3).

The paper contributes (1) a definition of CLI-first development (Section 3); (2) a single-product case study of the rule in practice (Section 4); (3) a caching-aware cost model that separates what an interface costs an agent into description, discovery and composition (Section 5); (4) an exploratory experiment and three follow-up waves, 2,144 runs in all, with every run, transcript and analysis released (Sections 6 and 7); and (5) principles for agent-ready CLIs and controls for agents that act through them (Sections 9–10).

2 RELATED WORK

Delegation and oversight. The debate between direct manipulation and delegation to agents is older than large language models [31, 55]. Norman warned that delegation works only when people keep an accurate sense of what the agent is doing and can take control back [44], and Horvitz's mixed-initiative principles [21] and Amershi et al.'s guidelines for human-AI interaction [2] turn that concern into design advice. Work on automation describes how oversight degrades: the operator of a highly automated system is left with the hardest decisions and the least practice at them [8], the right level of automation differs by stage of a task [46], and trust has to be calibrated to reliability [28]. An agent narrows the person's gulf of execution and can widen their gulf of evaluation [24], and stating intent is itself hard for non-experts [42, 63]. F5 bears on both gulfs.

Agent-computer interfaces and tool use. SWE-agent showed that interface design changes agent performance: a purpose-built interface resolved 18.0% of SWE-bench Lite tasks [26] against 11.0% for a plain shell [61]. CodeAct found that acting through executable code outperformed JSON and text tool calls by up to 20% in success rate across 17 models [56], and InterCode and Terminal-Bench evaluate agents in shells [35, 62]. Documentation alone lets models use unfamiliar tools about as well as demonstrations do [23], and ReWOO and LLMCompiler cut model invocations by planning tool calls ahead [27, 60]. Translating language into shell commands is a long-studied problem [29].

GUI, API, CLI and MCP agents. GUI agents beat CLI agents on 440 desktop tasks until the CLI's missing capabilities were filled in [67], and take 2.7 to 4.3 times more steps than necessary [1]; API-first and hybrid agents are faster [30, 65, 66], and MCP tools raise computer-use agents' success [25]. Comparisons of MCP and CLIs disagree. Across seven scaffoldings, paired MCP-to-CLI cost ratios ranged from 0.43 to 29 [18]; in 120 runs MCP was 23% faster and 2.5% cheaper than an equivalent CLI [64]. Holding a backend fixed, Smithery's 756 runs against public APIs found native MCP ahead of a CLI with descriptions on GitHub's REST API in success (91.7% against 83.3%), median tokens (28.8k against 82.9k) and time (10.4 against 24.9 s) [32, 43], and Scale Labs' 50 long-horizon tasks found the CLI's advantage depended on model and task and mattered little for frontier models [22].

Practice and lineage. "Agent experience" has become a design concern [7, 40], and products increasingly give the UI and the agent one set of actions [10, 49]. For CLI input, named flags beat single JSON payloads on accuracy and cost [33, 48], and Mistral gives every interactive prompt a flag [37]. The architecture underneath CLI-first is old: composable programs [34], ports and adapters [13], command-line design guidance [50] and API usability research [39]; the decay of a shared interface is a form of architecture erosion [14, 47]. CLI-first differs from API-first in naming which client must work first and giving it the role of reference.

How this study differs. The two fixed-backend studies are closest to ours. We differ in three ways: (1) we vary what the agent knows

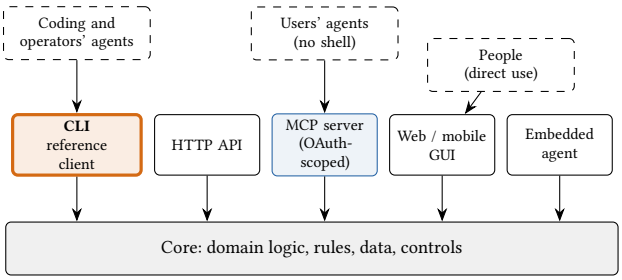


Figure 2: Every client calls one core; the CLI is the reference. Coding and operators' agents use the CLI; users' agents usually arrive through a scoped MCP server.

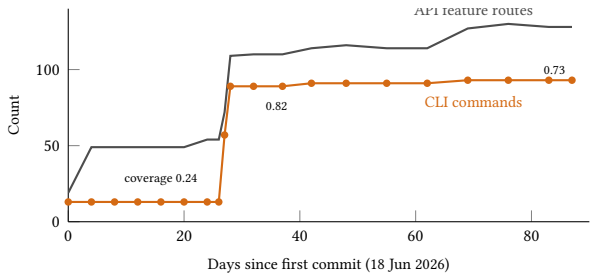


Figure 3: CLI coverage fell under a written rule until a dedicated phase restored it. Commands and API feature routes in Salli over its first 87 days; authentication routes are excluded.

about the interface and whether it may compose as separate factors, so their effects can be told apart; (2) we decompose cost by token type under prompt caching and model it, rather than reporting totals; and (3) we score a production product's database state rather than an agent's answer.

3 CLI-FIRST DEVELOPMENT

A team practises CLI-first development when it holds two rules. First, a capability is complete only when it can be exercised from the product's CLI; other clients may ship it at the same time or later, but not instead. Second, the CLI is the reference client: when the web application, the MCP server or an embedded assistant disagrees with the CLI about what the product does, the disagreement is a defect in one of them, and checks written against the CLI define the expected behaviour. The first rule without the second produces coverage that can drift; the second without the first produces a reference for part of the product.

Three kinds of agent interact with such a product (Figure 2). Coding agents and operators' agents run in a shell and are the CLI's main users: they exercise features without a browser, configure deployments and check their own changes. Embedded agents call the same core operations and get no more power than any other client. Users' agents, the assistants that end users delegate to, usually run where no shell exists and reach Salli through an MCP server with OAuth-scoped access. CLI-first serves them indirectly, by making every capability a scriptable operation that any protocol can expose; it does not guarantee that they will: Salli's open-source core has 125 CLI commands but 24 MCP tools.

The analogy with mobile-first [58, 59] holds for the order of work and fails on constraint: mobile-first started from the most limited client, CLI-first from the most capable, and its new user is an agent acting for a person. Accessibility is a closer precedent: the interfaces that expose an application to a screen reader stay complete only where teams treat gaps as defects.

4 A CLI-FIRST PRODUCT

Method. We studied Salli, a personal finance and tax product for Sri Lanka whose agent guidance adopts the CLI-first rule and which the author develops. The design is a single-case study with the author as participant-observer, in the tradition of autobiographical design [41] and following case-study guidance from software engineering [52]. The data are the product's git history (417 commits), architecture documents and commit messages at a fixed snapshot (13 September 2026). For CLI coverage we counted CLI commands and API route handlers, excluding authentication, at a series of dates; the ratio of counts is a proxy for parity, not a mapping of commands to routes. For agent involvement we counted commits whose message contains a `Co-Authored-By: Claude` trailer. The case can show what happened in one product; it cannot show how often CLI-first works elsewhere. Commit messages were themselves written with agent assistance and may be more explicit, and more favourable, than an independent record [9].

4.1 Coverage fell behind under a written rule and recovered in a dedicated phase

Salli combines a double-entry ledger, a versioned tax engine and AI agents that explain a user's financial position. Its domain core has no input/output or framework imports, and its CLI and HTTP API build their services through one composition function. The project's agent guidance states the order of work: "Core + CLI first — all domain logic runs from the terminal before any HTTP server exists," then the API, then web and mobile clients. 311 of its 417 commits carry an agent trailer.

The rule did not hold on its own. An HTTP API with 19 feature routes existed from the first day, and over four weeks the API grew while the CLI did not: by 14 July the CLI had 13 commands against 54 feature routes, a ratio of 0.24 (Figure 3). The roadmap then inserted a "CLI completion" phase ahead of new domain work, and within two days the ratio reached 0.82. Five domains built immediately afterwards each received core logic, API routes and a CLI command group within about twelve minutes, in one agent session. Coverage then fell to 0.73 as the API gained routes the CLI did not; the open-source core now runs a continuous-integration check that fails when a route has neither a command nor a recorded reason for lacking one. We have no data yet on whether the check holds coverage.

4.2 A pass through the CLI found a control that held on only one path

Once the CLI covered the product, a coding agent's pass through it found defects in the core. One agent-assisted

commit (`862ed06`, 19 July) records the defects "found during a full CLI end-to-end pass". The CLI had never set up the checkpointer that the HTTP server configures for the embedded agents, so the agents' history commands crashed and the human-approval gate on their write tools silently did nothing when invoked from the CLI. Data export crashed for any user who had computed their tax, and a command that posts imported transactions misreported how many it had posted. The approval-gate defect could not be seen from the web client, which reached the agents through the HTTP server that set the checkpointer up. That gate had been fixed for the web path a month earlier; the CLI pass found that the same guarantee did not hold on another path into the core. These observations, from one product and one participant-observer, motivate study S2 in Section 12.

5 WHAT AN INTERFACE COSTS AN AGENT

An MCP server publishes typed tools whose definitions the agent's host loads into the model's context [38]; the model calls tools one response at a time, possibly several in parallel. With a CLI that emits structured output, an agent can instead write a chain of operations once as a program and run it as one tool call, so that intermediate data move through pipes or files rather than through the model, as in Figure 1.

A cost model. Consider k operations, each needing the result of the one before. Let S be the fixed prompt prefix (system prompt and interface description), u the user's request, a_i and r_i the tokens of the i-th call and its result, c the tokens of a pipeline script and f its filtered output. With direct calls there are $k+1$ model passes, and pass j reads $P_j = S + u + \sum_{i<j}(a_i + r_i)$. Without caching, the input tokens processed are

$$I_{\text{calls}} = (k+1)(S+u) + \sum_{i=1}^{k}(k+1-i)(a_i + r_i), \tag{1}$$

quadratic in k when results are of similar size, while a single pipeline needs two passes:

$$I_{\text{pipe}} = 2(S+u) + c + f. \tag{2}$$

With prompt caching, cache writes cost ω times the base input rate and cache reads ρ times; for one provider $\rho = 0.1$, ω is 1.25 for five-minute and 2 for one-hour entries, and output costs five times the input rate [6]. Direct calls then cost approximately $\omega P_1 + \rho \sum_{j \le k} P_j + \omega \sum_i (a_i + r_i)$ input-token equivalents and the pipeline $\omega(S+u+c+f) + \rho(S+u)$ (Table 1). Caching removes most of the quadratic term but not the $k+1$ sequential passes, nor the cost of every intermediate result entering the context once. When many agents share a prefix, even the first pass may read S from the cache.

Discovery and repetition. Two further terms decide whether a pipeline is available. The first is *discovery*. Equation 2 assumes an agent that knows the commands and the shape of their output. An MCP server supplies that knowledge in S; an agent that must learn a CLI reads help instead, taking d extra passes that add h tokens, written to the cache once and re-read on every later pass. Whether this exceeds the CLI's saving in S depends on d, h and the passes that follow. The second is *repetition*. For n independent operations of one kind, direct calls can run in parallel within one pass, but the model writes all n calls and all n results enter the context. A loop over a CLI command avoids the per-call overhead; its length is constant in n when it iterates over the product's own data, as in Figure 1, and grows with n when the arguments must be written out.

Table 1: A pipeline needs two passes where direct calls need k+1. Input-token equivalents for $k = 5$ dependent operations with $S = 10$k, $u = 1$k, $a_i = 100$, $r_i = 2$k, $c = 400$, $f = 200$, $\omega = 1.25$, $\rho = 0.1$. The last row drops tool definitions from the prefix. The table uses the five-minute cache rate for illustration; the experiment used one-hour entries ($\omega = 2$).

	Passes	Uncached	Cached
Direct tool calls	6	97.5k	34.5k
CLI pipeline	2	22.6k	15.6k
CLI pipeline, $S = 1$k	2	4.6k	3.5k

Research questions. The model raises four questions. We formed the predictions after the first wave of the experiment (Section 6.2) and test them on data that include that wave, so they are exploratory.

- **RQ1, knowledge.** Does an agent that must learn a CLI from its help cost more than one given an MCP server's schemas? *Predicted: yes.*
- **RQ2, description size.** For the same operations, does an MCP server generated from the CLI cost more than a curated one? *Predicted: yes.*
- **RQ3, composition.** When the agent knows the commands, does a composed CLI cost less than the curated MCP server on chains of dependent operations, and on tasks that repeat an operation? *Predicted: yes, more so for repetition.*
- **RQ4, oversight.** What does composition do to what a person can review? *No prediction.*

Composition's economics come from code as the agent's action space, not from the shell; MCP tools can also be called from code in a sandbox [3, 4, 11], though standard MCP clients do not offer it. A CLI brings composition with the interface, along with costs: scripts must anticipate output shapes that MCP declares, chains can fail halfway, shell quoting is fragile [33], every invocation starts a process, and a shell carries ambient authority. A pipeline planned before reading data keeps untrusted content out of the control flow [15]; one that interpolates data into commands, like Figure 1's loop, does not.

6 AN EXPLORATORY EXPERIMENT

6.1 Design

We held one product, its data and its tasks fixed and varied the interface. The product was Salli's open-source core at commit `0277747`, whose CLI (125 commands, 120 with `--json`) and MCP server (24 tools) call the same services and database. A *configuration* fixes three factors (Table 2). The *protocol* is MCP tool calls or shell commands. *Knowledge* is what the interface description tells the agent before it starts. MCP configurations receive tool schemas. CLI configurations receive one of four descriptions: help alone;

Table 2: Eight configurations cross protocol, knowledge and composition. Knowledge gives the interface description's size (`o200k_base` tokens). 1st pass: median context of the first model call. Runs: retained runs (Haiku and Sonnet).

Configuration	Knowledge	Composition	1st pass	Runs
MCP-24	24 schemas (3.0k)	parallel calls	5.3k	96
MCP-102	102 schemas (9.7k)	parallel calls	17.2k	96
CLI skill/step	skill (0.6k)	one per call	3.8k	64
CLI skill/comp	skill (0.6k)	composed	3.9k	96
CLI help/comp	help only	composed	2.9k	64
CLI installed/comp	5 skills, on demand	composed	6.6k	96
CLI sheet/step	command sheet (1.1k)	one per call	4.7k	96
CLI sheet/comp	command sheet (1.1k)	composed	4.7k	96

the core's agent skill, pasted into the prompt (646 tokens naming command groups and rules, but no options); the core's five skills, installed for the agent to load if it chooses; or a *command sheet* generated from the CLI's own definitions (1,134 tokens giving usage, options and JSON output shape for the 14 commands that do what MCP-24's tools do). *Composition* is whether the agent may combine commands: one-per-call configurations permitted only the `salli` executable, while composed ones asked the agent to compose dependent steps and allowed pipes, `jq`, scripts and (from wave 4) Python. MCP-24 is the core's own, curated server, whose results are compact summaries. MCP-102 is generated from the CLI, with one tool for each of 102 of the 120 commands that support `--json` (administrative command groups are left out), and returns the CLI's JSON, so it differs from MCP-24 in both catalogue size and result size.

What the experiment can and cannot show. It compares these configurations on one product, six tasks and two models from one provider. It cannot say how often CLIs beat MCP servers in general, and it does not test the process claim of Section 4.

Tasks and scoring. Six tasks could all be completed with the operations available to MCP-24. Four are *chains* of $k = 1, 2, 4$ and 8 operations, with nominal dependency depths 1, 2, 4 and 6: report the tax owed; record a cash expense; record a salary, compute the tax and set a reminder carrying the figure; and create an account, post to it, find the expense account with the largest balance, refund it, and set a reminder carrying the tax and the bank balance. Two are *repetition tasks*: record 25 expenses listed in the request (bulk entry), and set a reminder for every expense account over LKR 20,000 (fan-out, Figure 1). Each ran on a small ledger (16 accounts, 30 entries) and a large one (106 accounts, 440 entries); fan-out needs one write on the small ledger and 24 on the large, which makes the small cell a control where composition has nothing to compose. A *cell* is one task on one ledger, so there are 12 cells and 24 combinations of model and cell. Every run used a freshly seeded user. The one-operation tax question was scored from the answer and the other tasks against the database, by the expected records; the two-step and repetition tasks also required that no other records of the same kind exist on the task's dates, while the 4- and 8-step checks did not look for extra records.

Interface asymmetries. MCP-24's reminder tool takes free text, while the CLI's `reminders add` takes it as a `KIND` argument that neither the skill nor the sheet explains. The CLI accepts account codes. The core's documentation, our prompt for the CLI configurations and the command sheet's header all said identifiers could be shortened to a unique prefix, which `entry add` did not accept. On the small ledger the 8-step task's target account is necessarily the one created in its first step.

Harness. The agent was Claude Code 2.1.293 in headless mode driving Claude Haiku 5.5 and Claude Sonnet 5.5. Its system prompt was replaced by a short task prompt common to all configurations plus a configuration-specific paragraph; user settings were excluded, MCP tool search disabled, and only the configuration's tools enabled. The composed configurations' paragraph alone gave a reason to compose, so "that intermediate output does not have to pass through you"; no other configuration was told anything about efficiency. The common prompt also said that the request was the user's confirmation for any change it asked for, so no configuration paused for approval. MCP-24 ran the core's server over stdio with its OAuth user lookup replaced by the seeded user. Composed configurations ran with permissions bypassed under a deny-list of deletion, network, version-control and package-manager commands, and with a directory holding only `salli` prepended to the path; there was no sandbox, and the environment held the test instance's database URL and service keys. No transcript shows a command that bypassed the CLI. Caching used one-hour entries shared across runs: in 702 of 704 runs the first call read the prompt and interface description from a cache warmed by earlier runs.

6.2 Collection history

The design grew in five waves, each started after the previous one's results were visible (Table 3). The first wave, a pilot on the chain tasks, found the CLI more expensive than MCP-24; the command-sheet configurations, the repetition tasks and the research questions came afterwards, and the pilot's 128 MCP-24 and CLI skill/step runs are part of the analysed data. Of 947 runs attempted, 243 were superseded and 704 retained; the run ledger lists each with its wave, parallelism, permissions and fate. Four harness problems caused the supersessions. Claude Code's allowlist matcher refused composed commands containing variables, quoting or loops (in 29 of 128 wave-1 composed runs, all of which still succeeded), so composed configurations were re-collected without it (128 runs). A deny-list that also blocked interpreters was in force for 32 repetition-task runs, three of which had a script refused; all 32 were re-collected. The composed chain runs we kept used that deny-list, and none invoked an interpreter. Parallel runs shared a temporary directory and overwrote each other's files, so the 80 composed runs that had used a shared path, and one that ran beside a smoke test, were collected again one at a time; because this rule selected runs by behaviour, Section 6.10 restores them. One run was stopped by the operator in error and one produced no output; both were collected again.

6.3 Measures and analysis

For each run we recorded success; cost; tokens of each kind, summed over the run's agent invocations; passes, tool calls and characters of tool output; and time as reported by the agent harness, summed over invocations. Cost is at list prices: $0.10,

Table 3: The design grew over five waves. Parallel: concurrent runs. Retained: runs in the analysis.

Wave	Content	Parallel	Retained
1	Pilot: MCP-24, CLI skill/step, skill/comp, help/comp; chains	6	128
2	Command-sheet configurations; composed configurations re-collected	8	224
3	CLI installed/comp	8	55
4	MCP-102 on chains; repetition tasks for six configurations	8	217
5	Shared-directory runs, re-collected one at a time	1	80

0.20, 0.01 and 0.50 per million uncached input, cache-write, cache-read and output tokens for Haiku, and $2, 4, 0.20 and 10 for Sonnet. These prices reproduce the harness's reported cost for every run. The outcome is cost relative to MCP-24: in each cell the ratio of geometric-mean costs, combined over cells by the geometric mean. We report two targets of inference. *For these tasks*, the 95% interval is a Welch-type t interval for the mean log ratio over cells, with runs within cells as the sampling unit and Satterthwaite degrees of freedom; p-values come from a stratified permutation test (10,000 permutations), Holm-adjusted over 24 comparisons [20]. A bootstrap over runs within cells [17] gives similar intervals and is in the released data. *For tasks like these*, a t interval over tasks uses each task's log ratio averaged over its two ledgers; with four chain tasks it is wide, and with two repetition tasks uninformative. We compare models as a ratio of ratios, overall and per task, and give absolute differences in cents.

6.4 Results

Success was at ceiling, so cost decides. Of 704 runs, 693 (98.4%) succeeded; per configuration and model, success ranged from 57/60 to 60/60 with overlapping Wilson intervals. Figure 4 and Table 4 give every cost effect; Table 5 gives the median cost per run that anchors them, from 0.05 cents (Haiku, MCP-24, one operation) to 70.9 cents (Sonnet, CLI sheet/step, bulk entry on the large ledger).

6.5 F1. Agents that learned the CLI from its help paid 1.6 to 2 times as much

An agent that had to learn the CLI from its help cost more than an agent given MCP-24's schemas (RQ1). On chains CLI help/comp cost 1.99 times as much with Haiku and 1.62 times with Sonnet, about 0.2 and 1.7 cents more per task, with intervals above 1 for these tasks and for tasks like these; it cost more in all 8 cells for Haiku and 7 of 8 for Sonnet. These agents made two to four help calls per run, and at $k = 8$ took a median of 15.5 passes against 7 for MCP-24 (Haiku). A skill that named the command groups but no options did not close the gap (1.22 to 2.08).

6.6 F2. The generated MCP server cost more than the curated one in every cell

MCP-102 cost more than MCP-24 in all 12 cells for both models (RQ2): 1.79 and 1.57 times as much on chains with Haiku and Sonnet, and 1.53 and 1.34 times on repetition tasks ($p = 0.002$ throughout). Its schemas took 9.7k tokens against 3.0k, and it also returned 2.3 to 3.4 times as many characters of results, so the comparison does not isolate catalogue size. Against the CLI it was generated from, used with the command sheet and composed, it was the more expensive interface on both models and both task families.

6.7 F3. On chains, the CLI with the command sheet could not be told apart from MCP for Sonnet, and cost more for Haiku

With the command sheet and composition, Sonnet's agents cost about what MCP-24's did on chains, and Haiku's cost more (RQ3). Sonnet's ratio was 0.89 (95% CI 0.78–1.02; Holm $p = 0.26$), under a cent less per task; over tasks like these the interval, 0.57–1.39, is too wide to claim either parity or a saving. Haiku's ratio was 1.21 (1.11–1.31; Holm $p = 0.002$), about 0.07 cents more per task. The models differed: Sonnet's ratio was 0.74 times Haiku's (0.67–0.82). At $k = 8$ on the large ledger Sonnet composed its work into three tool calls and four passes, and Haiku composed its work into ten calls and nine passes.

Result size is a third factor. The CLI's JSON is fuller than MCP-24's summaries. With one command per call, CLI sheet/step agents received about twice as many characters of results as MCP-24's (1.98 times for Haiku, 1.96 for Sonnet) and cost 1.28 and 1.24 times as much, although, like MCP-24's agents, they issued one operation per call. Composed agents could filter results before they reached the model, and received 1.64 and 1.07 times as many characters.

6.8 F4. Composition paid when an operation repeated

On the two repetition tasks, the composed CLI with the command sheet cost 36–39% less than MCP-24 (RQ3): 0.64 with Haiku (95% CI 0.56–0.73; Holm $p = 0.005$), about 0.3 cents less per task, and 0.61 with Sonnet (0.50–0.74; $p = 0.002$), about 4 cents less. Cost fell in three of four cells for Haiku and four of four for Sonnet. **The saving was not spread alike.** Haiku saved on bulk entry only (0.41; fan-out 1.02), while Sonnet saved on both (0.67 and 0.56). Sonnet's ratio was 1.65 times Haiku's on bulk entry and 0.55 times on fan-out, so the overall ratio of ratios, 0.95, hides an interaction between model and task. In the control cell, the fan-out that needs one write, composition had nothing to compose and gave no consistent advantage: MCP-24 was cheaper for Haiku (0.11 against 0.18 cents) and more expensive for Sonnet (1.83 against 1.61). Two tasks cannot establish generality; the interval over tasks spans 0.20 to 1.91 for Sonnet.

Composition without the sheet had similar point estimates that were not established (CLI skill/comp, 0.79 for Haiku and 0.64 for Sonnet; Holm $p = 0.19$ and 0.067), and restoring the superseded runs moves Sonnet's to 0.81. One command per call reversed the effect for Sonnet, at 2.90 times MCP-24. On bulk entry its agents issued the 25 writes in parallel, with account identifiers shortened to a prefix as the documentation, our prompt and the command sheet said they could be; `entry add` rejected each with a traceback, the agents repeated the writes, and the two bulk cells cost 5.7 and 6.6 times MCP-24.

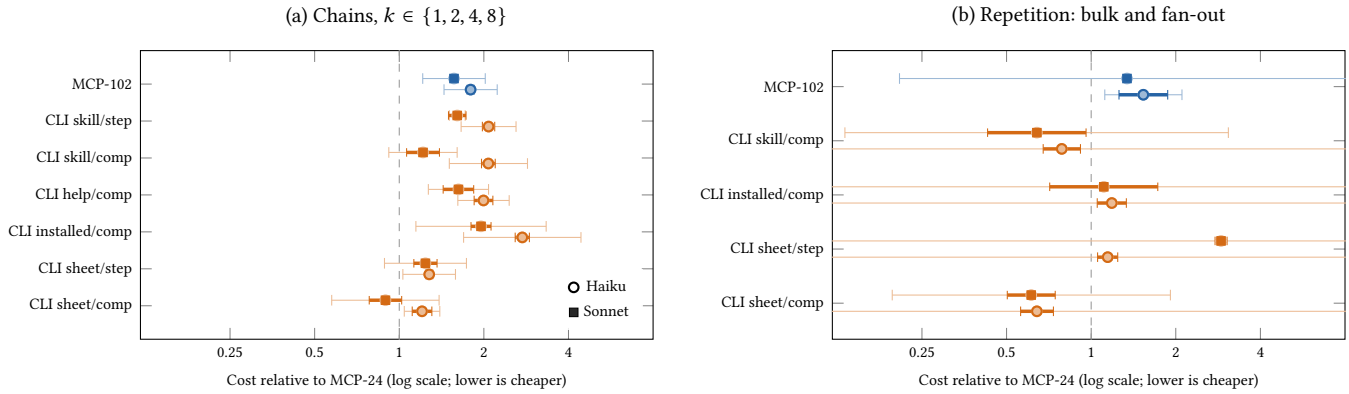


Figure 4: Knowledge, more than protocol, decides cost; composition pays when work repeats. Cost of each configuration relative to MCP-24 (dashed line), geometric mean over cells; blue: MCP, orange: CLI; hollow: Haiku, filled: Sonnet. Thick bars: 95% Welch intervals for these tasks. Thin bars: 95% intervals over tasks, clipped to the axis for the two repetition tasks. CLI help/comp and CLI skill/step were not run on repetition tasks.

Table 4: Without the command sheet, every CLI configuration cost more than MCP-24 on chains; with the sheet and composition, the CLI cost less on repetition tasks. Ratio: geometric mean over cells (lower is cheaper). For these tasks: 95% Welch t interval, runs within cells. For tasks like these: 95% t interval over tasks. p: stratified permutation test, Holm-adjusted over the 24 comparisons in this table; the smallest attainable value is 0.002.

	Haiku 5.5				Sonnet 5.5			
Configuration	ratio	these tasks	tasks like these	p	ratio	these tasks	tasks like these	p
Chains (4 tasks, 8 cells)								
MCP-102	1.79	1.74–1.85	1.44–2.23	0.002	1.57	1.53–1.60	1.21–2.02	0.002
CLI skill/step	2.08	1.98–2.19	1.66–2.60	0.002	1.61	1.50–1.73	1.51–1.72	0.002
CLI skill/comp	2.08	1.96–2.19	1.51–2.86	0.002	1.22	1.06–1.39	0.92–1.61	0.061
CLI help/comp	1.99	1.85–2.15	1.62–2.46	0.002	1.62	1.43–1.84	1.27–2.08	0.002
CLI installed/comp	2.74	2.58–2.90	1.69–4.43	0.002	1.95	1.80–2.12	1.15–3.33	0.002
CLI sheet/step	1.28	1.24–1.32	1.03–1.58	0.002	1.24	1.13–1.36	0.89–1.73	0.002
CLI sheet/comp	1.21	1.11–1.31	1.04–1.39	0.002	0.89	0.78–1.02	0.57–1.39	0.257
Repetition (2 tasks, 4 cells)								
MCP-102	1.53	1.26–1.87	1.12–2.10	0.002	1.34	1.31–1.37	0.21–8.62	0.002
CLI skill/comp	0.79	0.68–0.92	<0.1–>10	0.186	0.64	0.43–0.96	0.13–3.08	0.067
CLI installed/comp	1.18	1.05–1.33	<0.1–>10	0.282	1.11	0.71–1.73	<0.1–>10	0.605
CLI sheet/step	1.15	1.05–1.24	<0.1–>10	0.061	2.90	2.76–3.04	<0.1–>10	0.002
CLI sheet/comp	0.64	0.56–0.73	<0.1–>10	0.005	0.61	0.50–0.74	0.20–1.91	0.002

6.9 F5. Composition coarsened oversight, and neither protocol made reports reliable

The configuration that saved most put 25 writes behind one tool call (RQ4). With MCP-24 or one command per call, each tool call carried at most one write; on the bulk task the composed agents placed a median of 25 writes in a single call. No person approved anything in the experiment, so this describes what a reviewer would be shown, not how reviewers would fare: per-action approval would become approval of a script. In 31 of the 96 composed repetition runs the script also sent the writes' output to `/dev/null`, so not even the transcript records what each write returned. **Five of the 11 failures ended in a report of success**, 5 of 704 runs, as coded by the author. Two were with MCP-24 (an omitted account; the wrong one of two similar names), one each with MCP-102 and CLI sheet/step (each refunded the wrong account), and one with CLI skill/comp (reminder texts carrying stray quotation marks). One MCP-24 agent ended: "Created 23 reminders … for the expense accounts over LKR 20,000", when 24 qualified. The other six failures disclosed what had gone wrong.

6.10 Composition saved tokens rather than passes, and the conclusions survive a cold cache

Composition saved tokens rather than passes. Across the 24 combinations of model and cell, CLI sheet/comp took more passes than MCP-24 in 8, fewer in 4 and as many in 12; its saving came from what each pass carried. Table 6 decomposes two Sonnet cells. On the 8-step chain, CLI sheet/comp read 36k cached tokens

Table 5: Absolute cost anchors the ratios. Median cost per run in US cents at list prices (lower is cheaper), with each configuration's ratio to MCP-24. (a) Chains by number of operations k, pooled over ledgers. (b) Repetition tasks by ledger; Fan S is the one-write control.

(a) Chains	Haiku 5.5					Sonnet 5.5				
Configuration	k=1	2	4	8	ratio	k=1	2	4	8	ratio
MCP-24	0.05	0.15	0.24	0.44	1.00	1.15	2.89	5.21	7.03	1.00
MCP-102	0.12	0.29	0.40	0.71	1.79	1.52	5.51	7.28	11.00	1.57
CLI skill/step	0.13	0.29	0.57	0.88	2.08	1.73	5.09	8.77	11.33	1.61
CLI skill/comp	0.15	0.33	0.41	0.74	2.08	1.82	2.34	7.49	6.46	1.22
CLI help/comp	0.13	0.25	0.42	0.79	1.99	2.03	5.57	8.07	9.30	1.62
CLI installed/comp	0.21	0.36	0.63	0.94	2.74	2.96	7.85	7.79	8.74	1.95
CLI sheet/step	0.06	0.26	0.34	0.58	1.28	1.07	4.87	7.42	9.46	1.24
CLI sheet/comp	0.06	0.15	0.23	0.59	1.21	1.07	3.62	3.49	4.99	0.89
(b) Repetition	Bulk S	Bulk L	Fan S	Fan L	ratio	Bulk S	Bulk L	Fan S	Fan L	ratio
MCP-24	0.60	0.76	0.11	0.71	1.00	9.91	12.52	1.83	11.70	1.00
MCP-102	1.51	0.90	0.20	0.94	1.53	10.46	16.05	3.10	16.95	1.34
CLI skill/comp	0.32	0.36	0.22	0.53	0.79	8.11	4.20	2.60	5.10	0.64
CLI installed/comp	0.41	0.72	0.38	0.73	1.18	5.77	9.38	5.49	8.40	1.11
CLI sheet/step	0.56	0.86	0.16	0.99	1.15	64.90	70.91	2.46	15.86	2.90
CLI sheet/comp	0.23	0.30	0.18	0.47	0.64	6.91	8.64	1.61	4.40	0.61

Table 6: The composed agent wrote less and re-read less. Median tokens (thousands), passes, tool calls and cost (cents) in two Sonnet cells on the large ledger.

Config.	Cache write	Cache read	Out	Passes	Calls	Cost
Sonnet, $k = 8$ chain, large ledger						
MCP-24	16.8	63.9	1.3	5	8	9.3
MCP-102	27.7	193.5	1.4	6	9	16.3
CLI help/comp	15.9	128.4	1.7	11	10	11.0
CLI skill/comp	11.1	108.9	1.9	11	10	8.8
CLI sheet/comp	10.9	36.4	1.0	4	3	6.1
Sonnet, fan-out to 24, large ledger						
MCP-24	17.6	25.2	4.1	3	26	11.7
MCP-102	27.3	70.1	4.8	3	26	16.9
CLI skill/comp	8.0	32.9	1.1	5	4	5.1
CLI sheet/comp	7.7	17.2	0.9	3	2	4.4

against 64k for MCP-24, and three quarters of its saving came from smaller cache writes; the help-only agent took twice the passes and read twice as much. MCP-24 took 5 passes rather than $k + 1 = 9$ because it issued independent calls in parallel. On the fan-out, smaller cache writes (54%) and fewer output tokens (44%) made up the saving; on Haiku's bulk task output tokens made up 63%.

On chains, the command sheet mattered more than composition, and installed skills cost most. Holding composition fixed, replacing help with the sheet lowered the chain ratio from 1.99 to 1.21 (Haiku) and 1.62 to 0.89 (Sonnet); holding the sheet fixed, composition lowered it from 1.28 to 1.21 and 1.24 to 0.89. On repetition tasks composition did more, lowering the ratio from 1.15 to 0.64 and 2.90 to 0.61. Which part of the sheet mattered is the subject of the first follow-up (Section 7.1). Installed skills were loaded in 35 of 60 Haiku and 30 of 36 Sonnet runs, arrived as an extra pass, and made CLI installed/comp the most expensive configuration on chains (2.74 and 1.95).

The conclusions survive a cold cache and the superseded runs. Re-pricing every run as if its first pass had written the cache raises MCP-102's chain ratio to 2.44 and 2.38 and lowers CLI help/comp's to 1.41 and 1.17. CLI sheet/comp's ratios move by at most 0.11 and CLI sheet/step's by up to 0.32; two ratios that were near 1 and not significant move just below it (CLI skill/comp, Sonnet, chains, 1.22 to 0.99; CLI installed/comp, Sonnet, repetition, 1.11 to 0.97). Restoring the superseded shared-directory runs changes no ratio by more than 0.08 except CLI skill/comp on Sonnet's repetition tasks, from 0.64 to 0.81.

The curated MCP server was fastest, and the CLI failed in ways MCP-24 did not. In time reported by the agent harness, MCP-24 took a median of 9.5 seconds per run on chains (8.6 for Haiku, 11.5 for Sonnet), against 15.3 to 39.3 for the CLI configurations and 16.3 for MCP-102, which like the CLI starts a Python process per call. On repetition tasks it took 26.5 seconds, against 60.3 to 69.2 for every other configuration. Runs executed six to eight at a time in waves 1–4 and one at a time in wave 5. Raw Python tracebacks of up to 9,400 characters entered the context in 48 of 512 CLI runs and 5 MCP-102 runs. In 30 of the CLI runs they followed identifiers shortened to a prefix, which `entry add` rejected although the documentation and our command sheet said it would accept them; in 3 they followed account codes, and in 11 values that the agents' scripts computed and the transcripts do not show. All 5 MCP-102 runs passed account codes, which that commit did not accept either. Outputs over the harness's 30 KB limit were truncated to a preview in 56 CLI runs. Five composed runs took 11.6 to 15.9 minutes: in two, a command the agent wrote waited on standard input until the harness moved it to the background; in two, the agent called the trial balance once per account; one we could not explain. MCP-24 met none of these problems.

7 THREE FOLLOW-UP WAVES

After the first version of this paper we ran three follow-up waves with the same harness, Claude Code version (2.1.293), models and statistics: 1,440 runs, of which 1,431 succeeded. Every run attempted was kept. Neither the questions nor the analyses were pre-registered, and in each wave the Sonnet runs were collected after the Haiku results had been seen. The first wave asks which parts of the command sheet mattered (Section 7.1). The second runs the paper's tasks on the product as rewritten since (Section 7.2). The third separates two of the rewrite's changes (Section 7.3). The first two waves overlapped on one machine, with a median of 6 to 10 agents running at once, so we do not compare their timings. The third ran alone, four runs at a time. A follow-up run ledger records each run's wave, start, end and concurrency.

7.1 F6. Which parts of the sheet mattered depended on the model

Design. We derived four variants from the command sheet used above, each keeping the sheet's header: one without the JSON output shapes (608 tokens); one without the options (915); one with names and one-line descriptions only, which removes both (389); and one with the full sheet plus usage and options for 97 more commands, 111 of the CLI's 125 (4,620 tokens; commands nested two levels deep were left out). The first three form a two-by-two design, options by output shapes, which we analyse as main effects. Each variant ran composed, on the paper's commit, tasks and ledgers, five times per cell with Haiku and three times with Sonnet. MCP-24 and the full sheet ran again in the same wave as references (576 runs, of which 572 succeeded). The wave reproduced the main experiment. Against MCP-24, the full sheet cost 1.19 and 0.90 times as much on chains for Haiku and Sonnet (previously 1.21 and 0.89), and 0.57 and 0.59 times on repetition tasks (previously 0.64 and 0.61). Each configuration's absolute cost was within 10% of its value in the main experiment.

For Haiku both the options and the output shapes lowered cost on chains; for Sonnet only the output shapes did (Table 7). Averaged over the other element, removing the options cost Haiku 1.24 times as much on chains (1.17–1.32; Holm $p < 0.001$ over the eight main effects) and removing the output shapes 1.10 times as much (1.03–1.17; $p = 0.02$). Without options, Haiku's agents read help, a median of one call against none, and took more passes. For Sonnet, removing the output shapes cost 1.19 times as much (1.05–1.34; $p = 0.046$), and removing the options made no detectable difference (0.92, 0.81–1.03). On repetition tasks, where composition did the work, neither element had a detectable effect for either model. Sonnet's comparisons are noisier than Haiku's: at this commit 5 to 9 of its 36 runs per sheet, the reference included, hit tracebacks from the identifier-prefix defect.

Whether the sheet's size mattered depended on the cache. Under the warm shared cache, the four-times-larger sheet cost about what the full sheet did (0.96 and 1.00 for Haiku, 0.95 and 1.30 for Sonnet; none detectable). Re-priced as if each run had written its own cache, it cost 1.27 to 1.59 times as much: its 3,500 extra tokens are written once per conversation at the cache-write rate, as the cost model predicts (Section 5). Size matters wherever the prefix is not shared, and the experiment's warm cache hides that. For the same reason the ablation does not settle F2. By the cost model, re-reading MCP-102's larger catalogue accounts for a median of 16% to 53% of its extra cost, depending on model and task family (chains or repetition tasks), so its catalogue and its larger results both contributed.

The output shapes also kept writes on the record. Without them, the composed agents more often sent their writes' output to `/dev/null`. This happened in 17 of 32 repetition runs without output shapes and 19 of 32 with names only, against 4 of 32 with the full sheet and 7 of 32 without options. An agent that does not know what a write returns is more likely to discard it, which leaves a reviewer less to check.

7.2 F7. On the rewritten product, the composed CLI with a sheet cost less than MCP

Design. Salli's developer, the author, rewrote its core after the main experiment and finished the rewrite after this paper's first version, so the rewrite cannot be treated as independent of the study. At commit `1322f84` the CLI is a TypeScript client of the HTTP API. It names accounts by code or name as well as by id, adds filters such as `--type` and `--balances`, accepts the identifier prefixes its documentation promises, and reports errors without tracebacks. The MCP server, which we call MCP-46, grew from 24 to 46 tools, with 6,599 tokens of definitions, and still takes account ids only. We ported the harness, keeping the prompts, tasks, ledgers and checks. Each run's CLI was authenticated by a personal access token that was revoked afterwards. Scripted right and wrong solutions of every task through both protocols confirmed that the checks still tell them apart (48 cases); that the tasks are equally hard on both versions is not established. The command sheet and the agent skill were regenerated from the new CLI by the same rules. The sheet covers 33 commands in 3,010 tokens, so each configuration name in Table 8 denotes a regenerated description. Five configurations ran five times per cell with Haiku and three times with Sonnet (480 runs, of which 475 succeeded).

With the sheet and composition, the CLI now cost less than MCP-46 for both models on these tasks (Table 8). On chains it cost 0.90 (0.84–0.96) times as much for Haiku and 0.60 (0.49–0.74) for Sonnet, cheaper in 6 and 8 of 8 cells; on repetition tasks, 0.48 and 0.32 times. Relative to the main experiment these ratios changed by factors of 0.74 (0.67–0.82) and 0.67 (0.54–0.84) on chains and 0.76 and 0.52 on repetition tasks. Re-running the paper's commit in the ablation wave changed the same ratios by factors of only 0.89 to 1.01, every interval containing 1. Over tasks like these, the chain interval excludes 1 for Sonnet (0.53–0.68) but not for Haiku (0.76–1.06). The result holds when each run writes its own cache (0.81 and 0.67 on chains). A CLI learned from help still cost Haiku twice as much on chains under the warm cache (2.00, 1.91–2.09), unchanged from the main experiment. Re-priced with a cold cache, it cost 1.10 times as much for Haiku and 0.78 times for Sonnet, because MCP-46's larger catalogue is expensive to write.

Several differences between the versions line up with the change. Each of the following is consistent with it, and the next subsection tests the first two.

Table 7: Which parts of the command sheet mattered depended on the model. Cost of the composed CLI with each sheet, relative to the full sheet (lower is cheaper): geometric mean over cells, with a 95% Welch interval over runs. The first row compares the full sheet with MCP-24 in the same wave. The last block treats the first three variants as a two-by-two design and gives the effect of removing each element, averaged over the other. Bold: Holm-adjusted $p < 0.05$ within each block (4 comparisons in the first row, 16 for the variants, 8 for the main effects; the interaction is shown unadjusted). Tokens: `o200k_base`.

		Haiku 5.5		Sonnet 5.5	
Sheet	**Tokens**	Chains	Repetition	Chains	Repetition
Full sheet vs MCP-24	1,134	**1.19 (1.10–1.30)**	**0.57 (0.48–0.67)**	0.90 (0.75–1.08)	**0.59 (0.43–0.82)**
Full sheet (reference)	1,134	1	1	1	1
No output shapes	608	1.04 (0.98–1.11)	0.99 (0.87–1.12)	1.23 (1.02–1.48)	1.11 (0.73–1.69)
No options	915	1.18 (1.06–1.31)	1.05 (0.88–1.26)	0.94 (0.79–1.13)	0.99 (0.69–1.42)
Names only	389	**1.36 (1.26–1.48)**	1.05 (0.89–1.24)	1.09 (0.89–1.32)	1.21 (0.89–1.66)
Every command	4,620	0.96 (0.86–1.08)	1.00 (0.89–1.14)	0.95 (0.77–1.18)	1.30 (0.90–1.89)
Main effects, two-by-two design					
Removing the options		**1.24 (1.17–1.32)**	1.05 (0.94–1.18)	0.92 (0.81–1.03)	1.04 (0.82–1.31)
Removing the output shapes		**1.10 (1.03–1.17)**	0.99 (0.89–1.11)	**1.19 (1.05–1.34)**	1.17 (0.92–1.48)
Interaction (both removed)		1.11 (0.99–1.26)	1.00 (0.81–1.25)	0.94 (0.73–1.20)	1.10 (0.69–1.76)

Table 8: On the rewritten product the composed CLI with a sheet cost less than MCP-46, a change beyond run-to-run drift. Each row gives cost relative to that version's own MCP server (lower is cheaper), main experiment (`0277747`, MCP-24) → rewritten core (`1322f84`, MCP-46), with a 95% Welch interval over runs. "Change": the rewritten-core ratio relative to the main experiment's (a ratio of ratios), with its interval. The last row repeats this for the paper's commit, run again in the ablation wave. Bold: Holm-adjusted $p < 0.05$ (16 comparisons for the ratios, 14 for the changes). –: not run in the main experiment.

	Haiku 5.5		Sonnet 5.5	
Configuration	Chains	Repetition	Chains	Repetition
CLI help/comp	1.99 → **2.00 (1.91–2.09)**	– → 1.01 (0.94–1.08)	1.62 → **1.34 (1.08–1.66)**	– → **0.62 (0.58–0.66)**
change	×1.00 (0.92–1.09)	–	×0.82 (0.68–0.99)	–
CLI skill/comp	2.08 → **1.50 (1.42–1.58)**	0.79 → **0.72 (0.66–0.79)**	1.22 → 1.02 (0.83–1.24)	0.64 → **0.42 (0.26–0.69)**
change	**×0.72 (0.67–0.78)**	×0.92 (0.77–1.08)	×0.84 (0.69–1.01)	×0.65 (0.41–1.04)
CLI sheet/step	1.28 → **1.12 (1.05–1.20)**	1.15 → **0.77 (0.69–0.87)**	1.24 → 1.01 (0.83–1.23)	2.90 → **0.73 (0.71–0.77)**
change	**×0.88 (0.82–0.94)**	**×0.68 (0.59–0.77)**	×0.81 (0.67–0.99)	**×0.25 (0.24–0.27)**
CLI sheet/comp	1.21 → **0.90 (0.84–0.96)**	0.64 → **0.48 (0.44–0.53)**	0.89 → **0.60 (0.49–0.74)**	0.61 → **0.32 (0.29–0.34)**
change	**×0.74 (0.67–0.82)**	**×0.76 (0.65–0.89)**	**×0.67 (0.54–0.84)**	**×0.52 (0.43–0.63)**
Same commit re-run (drift)	×0.99 (0.89–1.11)	×0.89 (0.72–1.09)	×1.01 (0.83–1.24)	×0.97 (0.70–1.34)

- *The MCP server became more expensive.* Its absolute cost rose by 0 to 15%. That accounts for about half of Haiku's change on chains and up to a third of the others.
- *Only the CLI gained code and name addressing.* On the bulk task, the CLI's one-call-per-write agents passed the request's account codes in 800 of 800 writes, while MCP-46's agents looked up ids and passed them in 800 of 800.
- *The CLI's errors no longer carry tracebacks.* None appeared in any of 384 CLI runs, against 48 of 512 before. With one command per call, Sonnet's repetition ratio fell from 2.90 to 0.73.
- *The composed agents received less text than MCP-46's,* 0.21 to 0.88 times as many characters against 0.78 to 1.64 before. This was not because the CLI's records became compact: an unfiltered account list is still 1.9 times as long per account as MCP's (226 against 121 characters; 2.3 times at `0277747`). The agents asked for less, using the new filters and `jq`.

The regenerated descriptions had flaws of their own. The sheet documents `accounts list`'s output as a list, but with `--balances` the command returns an object of accounts and balances. A `jq` error followed in 11 of the 14 Haiku sheet runs that used the flag. The generator recorded one output shape per command and missed that a flag changes it. The regenerated skill tells agents to confirm before they write, against the harness's rule that the request is the user's confirmation. In two runs Haiku asked for a go-ahead before computing tax, and failed.

7.3 What the isolating runs show

Design. A third wave tested the first two of those differences directly. It ran alone on the machine, four runs at a time (384 runs, all of which succeeded). MCP-24 as rewritten is MCP-46 serving only the 24 tools of the paper's MCP-24, with `1322f84`'s schemas (3,122

Table 9: Part of the rewritten core's advantage came from its MCP server's growth; account codes did not matter to the composed CLI. The isolation wave, collected with nothing else running (lower is cheaper; 95% Welch interval over runs). First block: cost relative to MCP-46. Second block: relative to the same server serving only the 24 tools of the paper's MCP-24. Last block: the CLI with ids only relative to the CLI with codes allowed, and the CLI's change since `0277747`, measured against MCP-24 in both versions (a ratio of ratios). Bold: Holm-adjusted $p < 0.05$ within each block (12, 8, and 4 and 4 comparisons).

	Haiku 5.5		Sonnet 5.5	
Configuration	Chains	Repetition	Chains	Repetition
Relative to MCP-46				
MCP-24 as rewritten (24 tools)	**0.88 (0.81–0.96)**	1.02 (0.98–1.07)	0.89 (0.73–1.08)	**0.95 (0.93–0.97)**
CLI sheet/comp	**0.84 (0.76–0.92)**	**0.53 (0.46–0.60)**	**0.61 (0.49–0.76)**	**0.37 (0.21–0.65)**
CLI sheet/comp, ids only	**0.86 (0.78–0.94)**	**0.56 (0.50–0.62)**	**0.66 (0.52–0.82)**	**0.52 (0.42–0.65)**
Relative to MCP-24 as rewritten				
CLI sheet/comp	0.95 (0.89–1.02)	**0.52 (0.45–0.59)**	**0.69 (0.56–0.84)**	**0.39 (0.23–0.68)**
CLI sheet/comp, ids only	0.97 (0.93–1.02)	**0.55 (0.49–0.61)**	**0.74 (0.60–0.91)**	**0.55 (0.44–0.68)**
Ids only vs codes allowed	1.02 (0.95–1.10)	1.05 (0.90–1.24)	1.07 (0.85–1.35)	1.39 (0.89–2.19)
CLI sheet/comp vs MCP-24: change since `0277747`	**×0.79 (0.71–0.87)**	×0.81 (0.67–0.97)	×0.77 (0.62–0.96)	×0.64 (0.40–1.02)

tokens of definitions). The CLI with ids only is the composed CLI with the sheet, with its account arguments documented as ids and its `salli` wrapper refusing account codes and names. It emulates a CLI that addresses accounts as MCP does. MCP-46 and the CLI with the sheet ran again as references and reproduced the rewrite wave: the ratio between them changed by factors of 0.93 to 1.17, every interval containing 1.

The MCP server's growth accounted for part of the change, and addressing by code for none of it (Table 9). Serving only the original 24 tools made the server cheaper for Haiku on chains (0.88, 0.81–0.96) and for Sonnet on repetition tasks (0.95, 0.93–0.97); the other two differences were not detectable. Requiring ids made no detectable difference to the composed CLI: 1.02 to 1.07, except 1.39 on Sonnet's repetition tasks with an interval of 0.89 to 2.19. Its scripts looked the ids up with `jq` at little cost. Against a server with the original 24 tools, the composed CLI with the sheet was level for Haiku on chains (0.95, 0.89–1.02) and cost less elsewhere. It cost 0.52 times as much for Haiku on repetition tasks, and 0.69 and 0.39 times for Sonnet on chains and repetition tasks. Measured against MCP-24 in both versions, the CLI's ratio fell by a factor of 0.79 (0.71–0.87) for Haiku on chains. The other three changes, 0.64 to 0.81, did not survive the correction. Which of the remaining differences produced that change is not identified: the filters, the absent tracebacks or the regenerated sheet. The sheet's mis-documented flag recurred: a `jq` error followed in 21 of the 24 Haiku runs that used `--balances`.

7.4 Oversight and the harness in the follow-ups

Of the follow-ups' nine failures, three were silent false successes. Two agents refunded the wrong account and reported the refund done, one through MCP-24 and one through the CLI. Another created 22 of 24 reminders and reported one for each qualifying account. Both wrong refunds followed a trial balance that lists balances by account id alone: the agents joined 106 bare identifiers to the account list in their reasoning, named the wrong account, and reported success. Two failures were the consent requests above and four were disclosed; the isolation wave had none.

The harness gave the agents more than they needed. The composed configurations ran with permissions bypassed under a deny-list and without a sandbox. Until the isolation wave, the agents' environment also held the test server's database address and service keys, which the TypeScript CLI does not need. No agent read the environment. Agents used `/tmp` directly instead of their own temporary directory in 83 of the 1,440 follow-up runs. In 10 runs they found the files in which Claude Code saves large outputs with a wildcard spanning every project on the machine. In the main experiment the same pattern picked up another session's output in two runs, whose transcripts we redacted; the follow-ups' transcripts show no such leak. One agent wrote into the home directory. In four runs agents listed the machine's processes. In the isolation wave, two Sonnet agents checking whether their writes had finished saw other concurrent runs' command lines this way, including another run's prompt and data. Neither acted on them, but runs on one machine were not isolated from one another. None of this harmed the runs, but it is the ambient authority that control C5 addresses. The access tokens were not scoped either: at `1322f84` a personal access token acts with the user's full rights.

8 THREATS TO VALIDITY

Construct. Cost was measured at one provider's prices under a warm shared cache; the cold-start analysis bounds that effect. Scoring checked the expected records, and same-kind extras on three of the six tasks, not the whole state; every failure was also read by hand. In the follow-ups, scripted solutions showed that the checks still separate right from wrong on the rewritten core, not that its tasks are as hard, and the regenerated sheet and skill differ from the originals in content as well as in the product they describe. The command sheet was generated for the operations the tasks needed and is a best case for CLI knowledge, so we also report help-only and skill configurations. Claude Code ran with its default reasoning settings, so output tokens include any reasoning, and agents could read the machine's clock: in the follow-ups, one noticed that the real date differed from the

Table 10: Each finding, its evidence, and how far it reaches.

	Finding and evidence	Scope
F1	A CLI learned from help cost 1.62–1.99× MCP-24 on chains	Both models; holds over tasks
F2	MCP-102 cost more than MCP-24 in all 12 cells, both models	Confounded with result size
F3	Sheet + composition on chains: 1.21× (Haiku), 0.89× (Sonnet, 0.78–1.02)	Sonnet not distinguishable from MCP-24
F4	Sheet + composition on repetition: 0.64× (Haiku), 0.61× (Sonnet)	Two tasks; Haiku on bulk only
F5	25 writes per tool call; 5 of 11 failures reported as successes	Both protocols; no human reviewers
F6	Removing the sheet's options: 1.24× (Haiku, chains); its output shapes: 1.10× (Haiku), 1.19× (Sonnet); a 4× larger sheet: 1.27–1.59× with a cold cache	Chains; repetition tasks unaffected
F7	Rewritten product, sheet + composition vs its MCP server: 0.90/0.60× (chains), 0.48/0.32× (repetition); vs the server's original 24 tools: 0.95/0.69×, 0.52/0.39×	These six tasks; versions differ in several ways

prompt's. Failures were coded from the agents' final reports by one coder; the codes and the quoted reports are released. **Internal.** The design changed between waves, and three permission regimes and two degrees of parallelism are confounded with configuration and family; the run ledger records each run's conditions, and restoring the superseded runs changes one ratio materially. Only the composed configurations were given a reason to compose, and only the CLI configurations were told about prefix identifiers. The follow-ups were not pre-registered, their Sonnet runs were collected after their Haiku results had been seen, and the ablation and rewrite waves shared one machine; the follow-up ledger records each run's conditions. **External.** One product, one harness, two models from one provider, and tasks drawn from what MCP-24 can do. Salli's CLI is absent from training data, unlike `git` or `gh`; `CLI help/comp` approximates an unfamiliar CLI, and a familiar one is left to the confirmatory study (Section 12). **Statistical conclusion.** With three to five runs per cell and two to four tasks per family, we report intervals for these tasks and for tasks like these separately, correct for 24 comparisons in the main experiment and within each follow-up table, and treat the research questions as exploratory. The author built the product, tasks and harness, and rewrote the product between the main experiment and the follow-ups (see the positionality statement).

9 PRINCIPLES FOR AGENT-READY CLIS

Table 11 lists nine principles. Most restate good CLI design [50] and advice on writing tools for agents [5]; the evidence column separates prior studies, the case, the experiment and practitioner reports.

10 CONTROLS FOR AGENTS THAT ACT THROUGH A CLI

An agent with a shell, a person's credentials and access to untrusted content is exposed to indirect prompt injection [16, 19]; with network access it has all three elements of Willison's lethal trifecta: private data, untrusted content and a way to send data out [57]. A coding agent that reads content generated from users' documents while holding an operator's credentials fits this description, and OWASP names the risk excessive agency [45]. Rules therefore have to be enforced in the core. We propose five controls and report which ones Salli has.

C1, approval out of band. A destructive or irreversible command should create a pending action on the server and return its identifier; the action runs only after a person approves it on a separate channel. A `--yes` flag the agent can pass is not consent. F5 makes this pressing: approval should attach to a script's effects, not its text. Salli implements it only for return preparation, and its approval gate for the embedded agents once failed open on the CLI path (Section 4).

C2, agent identities with least privilege. Agents should hold short-lived, scoped, read-only-by-default credentials distinct from the person's. Salli scopes external agents through OAuth for MCP. At `0277747` its CLI did not authenticate at all, acting as whichever user its local configuration named; at `1322f84` it authenticates with personal access tokens, but they carry no scopes and act with the user's full rights. Credentials in environment variables or arguments are visible to child processes and transcripts.

C3, deterministic computation where correctness matters. Figures people act on should come from code, with the model explaining them, as in Salli's rule that the model never computes money or tax.

C4, attributable audit. Every action should be logged with the acting agent and the person it acted for. At both commits Salli logs MCP tools' actions but not those of the CLI, which at `1322f84` reaches the core through an API that does not audit them, so the surface we recommend for agents is the one that is not audited.

C5, containment below the shell. Agents that run CLIs should do so in sandboxes with restricted file-system and network access, using signed, pinned releases. Command allowlists sit at the wrong level for composition: the harness's matcher refused composed calls in 29 of 128 runs, and our own harness, which then bypassed permissions under a deny-list without a sandbox, is an example of what teams do instead. Its agents used `/tmp`, read the harness's session files and listed processes (§7.4). Composed scripts add partial failure without rollback, retries without idempotency keys, and data interpolated into commands, so writes should be idempotent and every rule enforced in the core.

11 DISCUSSION

What CLI-first does and does not buy. The experiment does not support the claim that a CLI is the cheaper way for any agent to use a product (Table 10). A curated MCP server was cheaper than a CLI the agent had to learn, and faster. Its agents met no tracebacks, truncated outputs or stalled commands, and each of its actions could be reviewed on its own. Since users' agents reach Salli through MCP, nothing here argues for moving them. What the evidence supports is narrower: a CLI with a compact command sheet cost about as much as the curated server on chains for Sonnet, less on two repetition tasks, and less than a server generated from the same commands. On the product as rewritten since, the composed CLI with a sheet cost less than its MCP server on these tasks for both models, partly because the server had doubled its catalogue; against the server's original 24 tools it was level for the weaker

Table 11: Nine principles for agent-ready CLIs, with the evidence for each: prior work (cited), the case (§4), the experiment (§6) or practitioner reports.

	Principle	Evidence and notes
P1	Cover the product, and check it	Gaps send agents back to the GUI [67]; coverage fell in Salli until a dedicated phase restored it, and is now checked in continuous integration (Figure 3).
P2	Never wait for input	A flag for every interactive prompt [37]. The experiment's stalls came from commands the agents wrote, not from Salli's prompts, so the evidence is practitioners'. Consent is the exception (C1).
P3	Compact, legible structured output	`--json` on commands that return data, accepted as input by related commands, with filters so an agent can ask for less [5]; every composed run depended on it. The CLI's records were about twice as long as MCP-24's, and one-per-call agents paid for it (F3). On the rewritten product, composed agents filtered and received 0.21–0.88 times MCP's result text (F7). Results keyed by bare identifiers led to wrong-account false successes; return names with identifiers (§7.4).
P4	Named flags for input	Flags beat single JSON payloads on accuracy and cost [33]; one study, five runs per model.
P5	A command sheet with options and output shapes, matching behaviour	Documentation substitutes for demonstrations [23]; a 1.1k-token sheet of usage, options and output shapes cost less than help alone, a skill without options or skills loaded on demand (F1, F3). Without its options or its output shapes the weaker model cost more on chains, and without its output shapes the stronger model did (F6). A sheet that promises what the CLI does not do is costly: the original sheet's prefix claim led to most tracebacks, and the regenerated one mis-documented a flag's output (F7).
P6	Errors that name the problem; never a traceback	Tracebacks of up to 9,400 characters entered the context in 48 of 512 CLI runs, 30 of them after identifiers that the documentation and command sheet said were accepted. The rewritten CLI reports errors without them, and none entered any of its 384 runs (F7).
P7	Preview before change	`--dry-run` or diff for commands that change state [50]; a preview should return a plan identifier to which approval (C1) can be bound.
P8	Safe under concurrent use, with batch operations	Agents issued up to 25 invocations of one command in parallel; credential refresh, writes and connections must tolerate it. A command that takes many records in one call would give one-per-call agents part of composition's saving.
P9	Built on the shared contract, described for how it is cached	One composition function serves Salli's CLI and API; an interface generated from the CLI cost more as MCP than the curated one (F2), and a larger description cost more where its prefix was not shared (F6).

model on chains and cheaper elsewhere (F7). The process claim, that a complete CLI lets coding agents build and verify the product, rests on the case and is the subject of study S2 in the agenda below.

Description is a design object. The largest effects came from what the agent was told, not from the protocol. On chains, the command sheet did more than composition; a skill that taught rules without options did little; and skills loaded on demand cost more than a description supplied up front. On repetition tasks, composition did most of the work. For MCP, the curated server's advantage over the generated one points the same way, though that comparison also varies result size. The ablation narrows this: which parts of a sheet a model needed depended on the model, and its size mattered where the prefix was not shared (F6). Both suggest treating the interface description as something to design and measure. The newest models may need less of it [22].

What MCP can do instead. Several of the CLI's advantages here are not specific to a shell. Code execution with MCP presents a server's tools as a code API, so an agent can compose calls and filter data before it reaches the model [3, 11]. Programmatic tool calling does the same within the model API, and tool search loads tool definitions on demand rather than up front [4], which addresses the cost of a large catalogue that is not cached (F2, F6). Tool use examples show usage that a schema cannot express, the role the sheet's options and output shapes played in F6. Advice on writing tools recommends concise response formats, filtering, and names rather than opaque identifiers [5], as P3 does. None of these was tested here; a confirmatory study should include them.

Delegation, oversight and access. Composition trades the granularity of oversight for cost, and on both protocols agents sometimes reported success they had not achieved, in the follow-ups as in the main experiment. A sheet without output shapes also made composed agents discard what their writes returned more often (F6). A CLI is not itself accessible to people [53], though people who delegate do not use it. Whether delegation through agents improves access, and how people review composed actions, is the subject of study S4.

Limits. We did not measure the cost of maintaining coverage. CLI checks cannot see layout, rendering or client-side state, and checks that put a language model in the loop are no more deterministic than the browser tests they replace.

12 RESEARCH AGENDA

Table 12 lists four studies; S1 and S3 extend RQ1–RQ3 of Section 5. For S1, a confirmatory version of the experiment should be preregistered with a task-level analysis and use about 20 chain and 30 to 50 repetition tasks generated from a specification, three repeats, at least three models from two providers and two harnesses, isolated caches, a frozen harness and randomised run order. It should include an MCP server whose tools match the command sheet one for one, MCP with code execution or programmatic tool calling,

tool search, a CLI agents already know (gh), tasks beyond the curated server's operations, and whole-state checks. Its cost in model calls is small; writing and verifying tasks is the constraint.

13 CONCLUSION

We have argued that a product's capabilities should first become usable from its command line, and that this CLI should be the reference client against which its other clients are checked. In one product, coverage fell behind until a dedicated phase restored it, and a pass through the CLI found an approval gate that worked on only one path into the core. Holding the product fixed, we found that a CLI is not cheaper for agents because it is a CLI. Cost depended mostly on what the agent knew. Composition paid when an operation repeated, and in the main experiment the curated MCP server was fastest and easier to review. Follow-ups showed that which parts of a sheet an agent needs depends on the model. On the rewritten product, the composed CLI with a sheet cost less than its MCP server, partly because the server had grown and not because the CLI accepted account codes. For a CLI-first product, the description an agent receives, a command sheet for the CLI and a curated catalogue for MCP, is part of the interface and should be designed and measured like the rest of it.

Positionality and competing interests. The author founded Leaf Monkey Labs, which develops Salli and offers a hosted version commercially, and proposed the CLI-first thesis. The author built the product, the tasks, the command sheet and the experimental harness, and rewrote the product between the main experiment and the follow-ups. The experiment's adverse findings for the CLI are reported in full.

Artifact availability. Salli's core is open source under the AGPL-3.0 at https://github.com/leafmonkeylabs/salli-core; the experiment used commit `0277747`. Its public history begins when the core was separated from the hosted product in October 2026, so Section 4 was measured on the earlier private history. The paper's source, the case-study scripts, the harness, the generated command sheet and MCP server, all 704 transcripts, the 243 superseded runs, the run ledger, the collection logs, the follow-ups' 1,440 runs with their run ledger and logs, the harness ported to commit `1322f84`, and the analysis are at https://github.com/leafmonkeylabs/cli-first-paper and archived at https://doi.org/10.5281/zenodo.23284581 (all versions).

Use of generative AI. Generative AI tools were used for literature search, repository analysis, the experimental harness and drafting, and to simulate peer review of earlier drafts; each simulated reviewer's claims were checked against the data before any change was made. The author checked every figure and quotation against its source and takes full responsibility for the content.

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Table 12: Four studies would test CLI-first beyond one product.

	Question	Design	Measures
S1	Does the interface change agent outcomes when the product is fixed?	Explored in §6; the confirmatory design above, with injections planted in intermediate data.	Success by whole-state diff, cost by token type, passes, latency, injection success.
S2	Is the CLI a better verification surface for agent-built software?	Agents implement features under CLI acceptance checks, browser end-to-end tests or unit tests only; defects seeded in the core and the GUI; several products.	Defects caught by layer, escaped defects, flake rate, cost, time.
S3	Which CLI design choices matter to agents?	The sheet's elements are ablated in §7.1. Remaining: P2–P8 on a fixed product, one at a time, and more models.	Success and cost, especially for small models.
S4	How do people fare when they delegate through agents?	A natural-language front end held constant while the agent's interface and approval granularity vary (per action, per script, out of band); scripted errors and false success reports injected; synthetic data on a sandboxed backend; ethics approval and pre-registration.	Completion, time, detection of injected errors and false reports, understanding of what was done, trust and workload.

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